

GLOBAL TB INSIGHTS 2024

Exploring the Tuberculosis Burden

OVERVIEW

Global TB Insights 2024 is a healthcare analytics project focused on understanding the global tuberculosis burden using WHO Global Health Observatory data. The project was designed as a complete analytics workflow rather than only a dashboard exercise: raw datasets were inspected and cleaned in Excel/Power Query, standardized and combined into a master dataset, analysed with Excel calculations and Pivot Tables, and then transformed into an interactive Power BI report.

The analysis is restricted to 2024 so that the dashboard provides a focused cross-sectional view of TB incidence, estimated cases, deaths, childhood TB, HIV-associated TB, MDR/RR-TB, treatment coverage, geographic regions, and World Bank income groups.

OBJECTIVES

The objective of this project is to analyse the global tuberculosis burden in 2024 using WHO data and transform the data into meaningful insights through data cleaning, analysis, and interactive visualization.

Specifically, the project aims to:

- Analyse TB incidence, estimated cases, deaths, treatment coverage, childhood TB, HIV-associated TB, and MDR/RR-TB.
- Compare the TB burden across countries, WHO regions, and World Bank income groups.
- Identify countries and regions with comparatively higher TB burden.
- Examine differences in TB incidence and treatment coverage across regions and countries.
- Use Excel and Power Query to clean, transform, and prepare the data for analysis.
- Use Power BI to create an interactive dashboard that presents the findings clearly and supports data-driven interpretation.

PROBLEM STATEMENT

Tuberculosis remains a significant global health concern, with its burden varying across countries and regions. Although WHO provides extensive TB-related data, the information is distributed across multiple indicators and datasets, making it difficult to obtain a clear and consolidated view of the global TB burden. Therefore, there is a need to analyse and

consolidate the 2024 TB data to identify differences in disease burden and treatment coverage across countries and regions and present these findings in a clear and accessible manner.

SOURCE OF THE DATA

The data used in this project was obtained from the World Health Organization (WHO) Global Health Observatory (GHO), specifically the TB-related datasets covering tuberculosis cases and deaths.

The source provides tuberculosis indicators reported across countries, WHO regions, and World Bank income groups. The downloaded datasets contain multiple TB indicators and reporting years. For this project, the data was processed and filtered to focus specifically on 2024.

Source: WHO Global Health Observatory (GHO) – Tuberculosis Data

Link: <https://www.who.int/data/gho/data/themes/topics/topic-details/GHO/cases-and-deaths>

Domain: Healthcare / Tuberculosis

Analysis Year: 2024

DATA PRE-PROCESSING USING EXCEL AND POWER QUERY

A. Data Pre-Processing Using Excel

Before moving into Power Query, the WHO datasets were initially inspected and prepared in Excel to understand their structure and identify potential data-quality issues.

1. Initial Data Inspection

- Reviewed all available TB indicator sheets and reference tables.
- Checked the number of rows and columns in each sheet.
- Examined column names, data types, reporting years, locations, and indicator information.
- Compared the common columns across the TB datasets to determine whether they could be combined.

2. Checking Empty Columns

Each sheet was checked for completely empty columns. Columns containing **100% blank values** were identified and removed as they did not contribute any information to the analysis.

3. Checking Missing Values

Missing values were investigated using Excel functions such as:

- COUNTBLANK()
- COUNTA()
- COUNT()

The missing-value checks showed that **Parent Location** and **Parent Location Code** contained blanks in some records.

4. Handling Missing Parent Location Values

The blanks in **Parent Location** and **Parent Location Code** were reviewed before deciding how to handle them. Since these records could represent observations where a parent location was not applicable, the records were retained rather than deleted.

Additional columns were created:

- **IF Parent Location**
- **IF Parent Location Code**

The IF() and ISBLANK() functions were used:

=IF(ISBLANK([Parent Location]),"Not Applicable",[Parent Location])

=IF(ISBLANK([Parent Location Code]),"Not Applicable",[Parent Location Code])

This allowed blank structural values to be represented as “**Not Applicable**” while retaining the original information wherever it existed.

Conditional Formatting was then applied to the cells containing “**Not Applicable**” so that these records could be easily identified and visually distinguished during data checking.

This helped us clearly separate **structural blanks where a parent location was not applicable** from other missing or unknown values.

5. Duplicate Check

Duplicate records were investigated using Excel's duplicate-checking functionality and by reviewing combinations of fields such as **Indicator, Location, Period, and ID**.

No duplicate records requiring removal were identified.

6. Checking Numerical Values

Numerical columns were reviewed for:

- Negative values

- Values stored as text
- Incorrect numerical formats
- Inconsistent decimal/whole-number formats

No negative values requiring correction were found.

7. Text Standardisation

Text fields were checked for unnecessary spaces and inconsistent formatting using the TRIM() function.

8. Reference Table Investigation

The WHO reference tables were examined to understand the codes used for countries, regions, global classifications, and World Bank income groups.

The **INDEX + MATCH** functions were used where required to retrieve corresponding information from the reference tables.

B. Data Transformation Using Power Query

After the initial Excel investigation, the datasets were brought into **Power Query** for structured transformation and preparation.

1. Loading the Individual Tables

The 10 TB indicator tables were imported into Power Query separately. Each query was reviewed to ensure that the required columns were available and that the tables could be combined based on their common structure.

2. Removing Unnecessary Empty Columns

The completely empty columns identified during the Excel inspection were removed from the Power Query tables as well.

This reduced unnecessary fields and kept the datasets focused on information required for the analysis.

3. Standardising Column Names

Column names were reviewed and standardised across the individual queries so that corresponding fields had consistent names.

Examples included standardising fields such as:

- ID
- Indicator
- Location
- Period
- Numeric Value
- Parent Location
- Parent Location Code
- Date
- Time-related fields

This was important because the tables needed a consistent structure before they could be appended.

4. Setting Appropriate Data Types

Data types were checked and corrected in Power Query.

For example:

- **Numeric Value** → Decimal Number
- **Time Dimension / Period** → Whole Number
- **Date** → Date/Date-Time as appropriate
- Text-based fields → Text

Correct data types ensured that the values could be properly filtered, aggregated, and analysed later.

5. Creating Imputed Parent Location Columns

The missing Parent Location information was also handled within Power Query to make the transformation reproducible.

The additional fields created were:

- **Imputed Parent Location**
- **Imputed Parent Location Code**

Blank values were replaced with “**Not Applicable**”, while existing values were retained.

This preserved records that did not have a country-level parent location instead of removing them from the analysis.

6. Checking the Structure of All 10 Tables

Before appending, the 10 indicator tables were checked for common fields and compatible structures.

The tables included indicators related to:

- TB incidence
- Estimated TB cases
- New and relapse TB cases
- TB deaths
- HIV-associated TB
- Childhood TB
- MDR/RR-TB
- TB treatment coverage

The HIV-positive TB incidence table was also reviewed and it was found that it did **not contain 2024 records**.

7. Appending the Tables

Once the individual queries had been cleaned and standardised, the 10 indicator tables were **appended in Power Query**.

Appending was used because the datasets represented similar types of observations with a common column structure. Instead of joining columns horizontally, the records from the individual tables were stacked vertically into one master table.

The resulting combined dataset was named:

TB_Master_Data

The initial appended master table contained approximately **41,056 records across 19 columns** before applying the final year filter.

8. Filtering the Master Dataset to 2024

Since the project focuses exclusively on 2024, the **Period/Time Dimension** was filtered to retain only records belonging to **2024**.

After applying the 2024 filter and final preparation, the analytical master dataset contained:

1,739 records × 21 columns

This became the main dataset used for the subsequent Excel analysis and Power BI dashboard.

9. Final Data Flow

The complete preprocessing workflow can therefore be summarised as:

WHO Raw Data → Excel Inspection → Data Quality Checks → Missing-Value Handling → Power Query Transformation → Standardisation → Append 10 TB Tables → 2024 Filter → TB_Master_Data → Excel Analysis → Power BI

This process provided a consistent and structured dataset for analysing the global tuberculosis burden in 2024.

C. Excel Analysis

After the data was cleaned, transformed, and filtered to 2024, Excel was used for preliminary analysis before developing the Power BI dashboard. Formulas and Pivot Tables were used to summarise the data, identify key values, and understand patterns across countries, regions, and World Bank income groups.

1. Formulas and Functions Used

The following Excel functions were used during data checking and analysis:

Function	Purpose
COUNTBLANK()	Checked the number of blank cells during data-quality assessment.
COUNTA()	Counted non-empty cells in selected ranges.
COUNT()	Counted numerical values in selected columns.
IF()	Applied conditional logic while handling missing values.
ISBLANK()	Identified blank cells, particularly in Parent Location fields.
TRIM()	Removed unnecessary spaces from text values.
INDEX() + MATCH()	Retrieved the corresponding country or location based on a matched value.
MAX()	Identified the highest value within an analysis range.
MIN()	Identified the lowest value within an analysis range.
SUMIFS()	Calculated totals for specific conditions, such as TB cases by World Bank income group.

Analysis Formulas

INDEX + MATCH:

=INDEX(A7:A203,MATCH(MAX(B7:B203),B7:B203,0))

This was used to identify the country corresponding to the highest value in the selected range.

SUMIFS:

=SUMIFS(TB_Master_Data[Numeric Value],TB_Master_Data[Source Data Set],"TB incidence cases",TB_Master_Data[Spatial Dimension],"WORLDBANKINCOMEGROUP",TB_Master_Data[Spatial Dimension Value Code],"WB_LI")

This was used to calculate TB cases for a specific World Bank income group.

2. Pivot Table Analysis

Pivot Tables were created to summarise the 2024 TB data and examine the distribution of tuberculosis across different geographic and classification levels.

The main analysis sheets included:

- **Global TB Indicator Summary** – summarised the major TB indicators for the global level.
- **Estimated TB Cases by Country** – compared estimated TB cases across countries and identified countries with higher and lower case estimates.
- **TB Cases by World Bank Income Group** – compared the estimated TB burden across different income groups.
- **TB Deaths by Country** – examined the distribution of TB deaths across countries.
- **TB Incidence by Region** – compared TB incidence rates across WHO regions.
- **TB Treatment Coverage** – compared treatment coverage across WHO regions.

3. Purpose of the Excel Analysis

The Excel analysis provided an initial understanding of the dataset before visualization in Power BI. It helped validate the prepared data, identify important patterns and high-burden areas, and determine which indicators and comparisons would be most useful for the final dashboard.

The results from this stage were used as a foundation for developing the interactive Power BI report.

DATA VISUALISATION USING POWER BI

After completing the Excel analysis and preparing the final **TB_Master_Data** dataset, Power BI was used to create an interactive dashboard for analysing the global tuberculosis burden in 2024.

The Power BI report was designed to move from a **global overview** to **country-level analysis** and finally to a deeper examination of TB burden and relationships between indicators.

1. Importing the Final Dataset

The final **TB_Master_Data** dataset containing **1,739 records and 21 columns** was loaded into Power BI.

The fields were reviewed to ensure that numerical, categorical, location, and indicator fields were correctly recognised and could be used for calculations and visualisations.

The main analytical fields included:

- Indicator
- Source Data Set
- Location
- Spatial Dimension
- Spatial Dimension Value Code
- Numeric Value
- Period
- Parent Location
- Parent Location Code
- WHO region and World Bank income-group classifications

2. Creating DAX Measures

DAX measures were created to calculate the major TB indicators required for the dashboard. Measures were preferred over manually calculated values so that the results could respond dynamically to filters and slicers.

The main measures created were:

(i) Total TB Cases

Total TB Cases = `CALCULATE(SUM('TB_Master_Data'[Numeric Value]),
'TB_Master_Data'[Source Data Set] = "TB Incidence Cases",
'TB_Master_Data'[Spatial Dimension] = "GLOBAL")`

This measure calculates the total estimated TB incidence cases.

(ii) Total TB Deaths

Total TB Deaths = `CALCULATE( SUM('TB_Master_Data'[Numeric Value]), 'TB_Master_Data'[Source Data Set] = "TB Death", 'TB_Master_Data'[Spatial Dimension] = "GLOBAL")`

This measure calculates TB deaths excluding HIV-associated deaths.

(iii) TB Incidence Rate

TB Incidence Rate = `CALCULATE( SUM('TB_Master_Data'[Numeric Value]), 'TB_Master_Data'[Source Data Set] = "TB Incidence", 'TB_Master_Data'[Spatial Dimension] = "GLOBAL")`

This measure is used to represent the TB incidence rate per 100,000 population.

(iv) TB Treatment Coverage

TB Treatment Coverage = `CALCULATE( SUM('TB_Master_Data'[Numeric Value]), 'TB_Master_Data'[Source Data Set] = "TB Treatment Coverage", 'TB_Master_Data'[Spatial Dimension] = "GLOBAL")`

This measure represents the percentage of TB cases covered by treatment.

(v) TB New and Relapse Cases

TB New and Relapse Cases = `CALCULATE( SUM('TB_Master_Data'[Numeric Value]), 'TB_Master_Data'[Source Data Set] = "TB New and Relapse", 'TB_Master_Data'[Spatial Dimension] = "GLOBAL")`

This measure calculates the number of new and relapse TB cases.

(vi) HIV Positive TB Cases

HIV Positive TB Cases = `CALCULATE( SUM('TB_Master_Data'[Numeric Value]), 'TB_Master_Data'[Source Data Set] = "TB HIV Pos Cases", 'TB_Master_Data'[Spatial Dimension] = "GLOBAL")`

This measure calculates TB incident cases among HIV-positive individuals.

(vii) HIV Negative TB Death Rate

HIV Negative TB Death Rate = `CALCULATE( SUM('TB_Master_Data'[Numeric Value]), 'TB_Master_Data'[Source Data Set] = "HIV Neg Deaths", 'TB_Master_Data'[Spatial Dimension] = "GLOBAL")`

This measure represents the TB death rate among HIV-negative people.

(viii) TB Cases in Children

TB Cases in Children = `CALCULATE( SUM('TB_Master_Data'[Numeric Value]), 'TB_Master_Data'[Source Data Set] = "TB Children", 'TB_Master_Data'[Spatial Dimension] = "GLOBAL")`

This measure calculates TB incidence cases among children aged 0–14.

(ix) Country TB Cases

Country TB Cases = `CALCULATE( SUM('TB_Master_Data'[Numeric Value]), 'TB_Master_Data'[Source Data Set] = "TB Incidence Cases", 'TB_Master_Data'[Spatial Dimension] = "COUNTRY")`

This measure is used for country-level TB case analysis.

(x) Country TB Incidence Rate

Country TB Incidence Rate = `CALCULATE( MAX('TB_Master_Data'[Numeric Value]), 'TB_Master_Data'[Source Data Set] = "TB Incidence", 'TB_Master_Data'[Spatial Dimension] = "COUNTRY")`

This measure is used to analyse TB incidence rates at the country level.

(xi) Country TB Treatment Coverage

Country TB Treatment Coverage = `CALCULATE( MAX('TB_Master_Data'[Numeric Value]), 'TB_Master_Data'[Source Data Set] = "TB Treatment Coverage", 'TB_Master_Data'[Spatial Dimension] = "COUNTRY")`

This measure is used to analyse treatment coverage at the country level.

(xii) Country HIV Positive TB Cases

Country HIV Positive TB Cases = `CALCULATE( SUM('TB_Master_Data'[Numeric Value]), 'TB_Master_Data'[Source Data Set] = "TB HIV Pos Cases", 'TB_Master_Data'[Spatial Dimension] = "COUNTRY")`

This measure is used to analyse HIV-positive TB cases by country.

3. Dashboard Theme and Design

A **Slate Navy** colour theme was selected for the report to maintain a consistent and professional appearance across all three pages.

The dashboard design focused on:

- Clear KPI presentation
- Consistent visual hierarchy
- Meaningful chart titles
- Appropriate axis titles
- Consistent number formatting
- Interactive filtering
- Easy comparison across geographic and income classifications

Dashboard 1 – Global Tuberculosis Overview

The first dashboard provides a high-level overview of the global TB situation in 2024.

The dashboard includes:

- **KPI cards** for major TB indicators
- **Map** showing the geographic distribution of TB burden
- **TB incidence by WHO region**
- **TB distribution by World Bank income group**
- **Top 10 countries by estimated TB cases**
- **Top 10 countries by TB deaths**



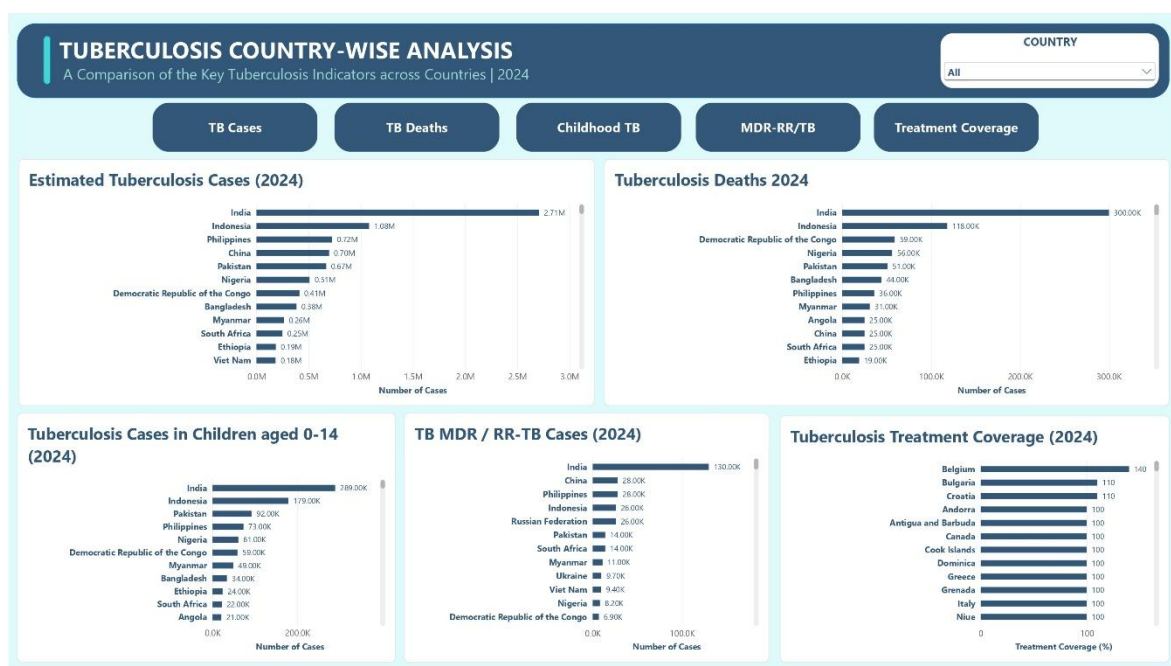
This dashboard was designed to allow users to understand the overall scale and geographical distribution of tuberculosis before moving into detailed analysis.

Dashboard 2 – Country-wise Analysis

The second dashboard focuses on country-level TB indicators.

The dashboard includes visualisations for:

- MDR/RR-TB incident cases
- Estimated TB cases
- TB treatment coverage
- TB cases among children aged 0–14
- TB deaths



Country-level filtering allows users to focus on individual countries and examine their TB burden across multiple indicators.

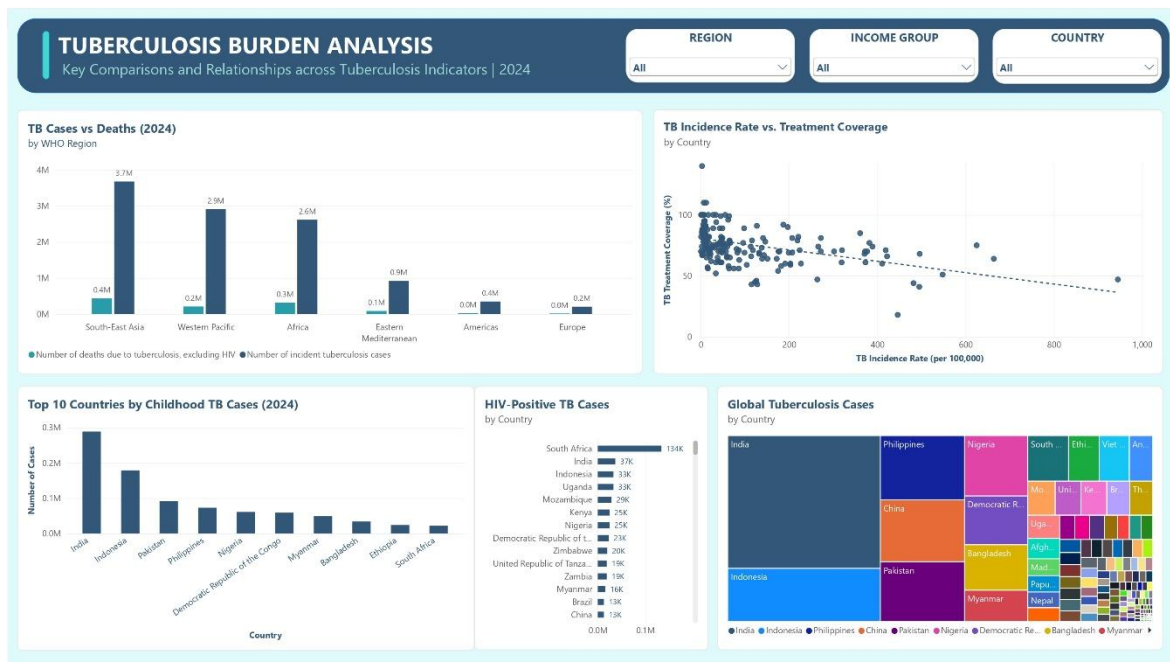
Dashboard 3 – Tuberculosis Burden Analysis

The third dashboard provides a deeper analytical view of the tuberculosis burden.

The visualisations include:

- HIV-positive TB cases by country
- TB cases versus TB deaths by WHO region
- TB incidence rate versus treatment coverage using a scatter chart
- Global TB cases using a treemap

- **Top 10 countries by childhood TB cases**



Slicers were included for:

- WHO Region
- Country
- World Bank Income Group

These allow users to explore different sections of the dataset interactively.

4. Interactive Features

Interactive features were incorporated throughout the report to allow users to explore the data dynamically.

The dashboard supports:

- Country selection
- WHO region filtering
- World Bank income-group filtering
- Cross-filtering between visuals
- Dynamic KPI updates
- Comparison of multiple TB indicators

The use of measures and interactive visuals allows the report to respond to user selections rather than presenting only static values.

5. Visualisation Approach

Different visualisation types were selected according to the nature of the analysis:

- **KPI cards** – for key global indicators
- **Map** – for geographical distribution
- **Bar charts** – for country and regional comparisons
- **Donut chart** – for income-group distribution
- **Scatter chart** – for examining the relationship between incidence rate and treatment coverage
- **Treemap** – for showing the relative distribution of global TB cases
- **Slicers** – for interactive filtering

The visuals were selected to make comparisons and patterns easier to understand rather than simply displaying the available data.

6. Final Power BI Report

The completed Power BI report brings together the cleaned WHO data, DAX measures, and visual analysis into a three-dashboard interactive dashboard.

The report follows the analytical flow:

Global Overview → Country-wise Analysis → Detailed TB Burden Analysis

This structure provides a clear progression from understanding the overall tuberculosis burden to exploring individual countries and relationships between key TB indicators.

FINDINGS AND INSIGHTS

The analysis of WHO tuberculosis data for 2024 revealed several important patterns in the global TB burden. The findings below were derived through Excel analysis, Pivot Tables, and the Power BI dashboard.

1. Global TB Burden

The analysis estimated a global TB incidence burden of approximately **10.70 million cases** in 2024, highlighting the continued scale of tuberculosis as a global health concern.

The estimated number of **TB deaths excluding HIV-associated deaths was approximately 1.08 million**, showing that TB continues to result in a substantial number of deaths worldwide.

2. Countries with the Highest Estimated TB Cases

The country-level analysis showed that TB burden was highly concentrated among a relatively small number of countries.

The countries with the highest estimated TB cases were:

1. **India – 2.71 million**
2. **Indonesia – 1.08 million**
3. **Philippines – 724,000**
4. **China – 696,000**
5. **Pakistan – 669,000**
6. **Nigeria – 510,000**
7. **Democratic Republic of the Congo – 412,000**
8. **Bangladesh – 384,000**
9. **Myanmar – 263,000**
10. **South Africa – 249,000**

India had the highest estimated TB burden among the countries analysed, with approximately **2.71 million estimated cases**.

3. Countries with the Highest TB Deaths

The analysis also identified the countries with the highest estimated TB deaths:

1. **India – 300,000**
2. **Indonesia – 118,000**
3. **Democratic Republic of the Congo – 59,000**
4. **Nigeria – 56,000**
5. **Pakistan – 51,000**
6. **Bangladesh – 44,000**
7. **Philippines – 36,000**
8. **Myanmar – 31,000**
9. **China – 25,000**
10. **South Africa – 25,000**

India recorded the highest number of estimated TB deaths, followed by Indonesia.

4. TB Incidence Across WHO Regions

The regional comparison showed considerable variation in TB incidence rates.

- **Africa – 207 per 100,000**
- **South-East Asia – 201 per 100,000**
- **Western Pacific – 131 per 100,000**
- **Eastern Mediterranean – 112 per 100,000**
- **Americas – 33 per 100,000**
- **Europe – 22 per 100,000**

Africa and South-East Asia recorded the highest incidence rates among the WHO regions analysed, while Europe had the lowest.

This demonstrates that the intensity of TB transmission varies substantially across regions.

5. TB Burden by World Bank Income Group

The analysis showed that the TB burden was not evenly distributed across income groups.

Estimated TB cases by World Bank income group were approximately:

- **Lower Middle Income – 6.49 million**
- **Upper Middle Income – 2.59 million**
- **Low Income – 1.24 million**
- **High Income – 168,000**

Lower-middle-income countries accounted for the largest share of the estimated TB burden.

This indicates that TB remains particularly significant in countries facing economic and healthcare-system challenges.

6. TB Treatment Coverage

Global TB treatment coverage was approximately **78%**.

Regional treatment coverage showed differences across WHO regions:

- **South-East Asia – 85%**
- **Europe – 79%**
- **Americas – 75%**
- **Western Pacific – 75%**

- **Africa – 74%**
- **Eastern Mediterranean – 72%**

South-East Asia recorded the highest treatment coverage among the regions analysed, while the Eastern Mediterranean recorded the lowest.

The comparison highlights differences in the extent to which identified TB cases are covered by treatment.

7. Childhood TB Burden

The analysis estimated approximately **1.19 million TB cases among children aged 0–14** globally.

India recorded the highest number of childhood TB cases among the countries analysed.

This highlights the importance of considering vulnerable population groups separately when assessing the overall TB burden.

8. HIV-Associated TB

The analysis estimated approximately **619,000 TB incident cases among HIV-positive individuals**.

At the country level, **South Africa recorded approximately 134,000 HIV-positive TB cases**, making it the highest among the countries highlighted in the analysis.

This demonstrates the important relationship between TB and HIV, particularly in countries with a significant HIV burden.

9. MDR/RR-TB Burden

The analysis estimated approximately **390,000 incident MDR/RR-TB cases** globally.

India recorded approximately 130,000 MDR/RR-TB incident cases, the highest among the countries highlighted in the analysis.

The presence of MDR/RR-TB adds complexity to TB control because drug-resistant cases require more specialised diagnosis and treatment.

10. New and Relapse TB Cases

The analysis identified approximately **8.33 million new and relapse TB cases**.

The regional analysis showed that **South-East Asia accounted for approximately 3.7 million incident cases**, indicating its substantial contribution to the global TB burden.

11. Overall Pattern in the Data

The analysis shows that the global TB burden is characterised by significant **geographical and economic disparities**.

A relatively small group of high-burden countries contributes a large proportion of global TB cases and deaths. At the same time, regions differ considerably in both TB incidence and treatment coverage.

The results also show that the TB burden extends beyond overall case counts, with important contributions from **childhood TB, HIV-associated TB, and MDR/RR-TB**.

12. Key Takeaway

The 2024 analysis demonstrates that tuberculosis remains a major global health challenge, but its burden is **not evenly distributed**. Countries, regions, and income groups experience substantially different levels of TB incidence, cases, deaths, and treatment coverage.

The combination of Excel analysis and the interactive Power BI dashboard makes these differences easier to identify and compare, providing a consolidated view of the **global tuberculosis landscape in 2024**.

RECOMMENDATIONS

Based on the findings from the 2024 analysis, the following areas can be considered for greater attention:

1. Prioritise High-Burden Countries and Regions

Countries and regions with comparatively high TB cases, deaths, and incidence rates should receive greater attention in TB prevention, detection, treatment, and monitoring efforts.

2. Strengthen Treatment Coverage

The differences in treatment coverage across WHO regions indicate a need to identify areas with comparatively lower coverage and improve access to timely TB diagnosis and treatment.

3. Give Greater Attention to Childhood TB

With approximately **1.19 million TB cases among children aged 0–14**, childhood TB should remain an important component of TB surveillance and control strategies.

4. Strengthen TB and HIV-Related Care

The presence of approximately **619,000 HIV-positive TB cases** highlights the importance of coordinated approaches to TB and HIV-related healthcare, particularly in countries with a higher burden of HIV-associated TB.

5. Address Drug-Resistant TB

The estimated **390,000 MDR/RR-TB cases** demonstrate the importance of continued focus on the detection, monitoring, and appropriate management of drug-resistant tuberculosis.

6. Focus on Lower-Income Settings

The concentration of TB cases within **lower-middle-income and low-income groups** suggests that strengthening healthcare capacity, access to diagnosis, and treatment services in these settings can be an important area of focus.

7. Use Data for Continuous Monitoring

Regular analysis of TB indicators across countries, regions, and income groups can help identify changes in disease burden, treatment coverage gaps, and emerging areas that require greater attention.